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Why Tanks Still Matter in Modern Warfare

SPOTLIGHT

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Military experts and strategic think tanks believe that achieving victory in war ultimately requires a physical ground presence. While drones, missiles, air power and cyber warfare can inflict heavy damage on military assets and infrastructure, they cannot occupy territory, enforce surrender or establish political control. History consistently shows that long-term military and political success requires ground forces.

The role of tanks in modern warfare has become a subject of intense debate. Critics argue that the proliferation of drones and precision-guided weapons has reduced their battlefield relevance, while supporters say that tanks remain indispensable as mobile armoured platforms capable of delivering overwhelming firepower, breaching enemy defences and supporting infantry.

A senior Indian Army officer said drones can create panic and inflict damage, but they cannot achieve the ultimate objective of war. "To enforce surrender, a ground invasion of enemy territory is essential. That is what India demonstrated during the 1965 and 1971 wars against Pakistan. Tanks remain critical for such operations," the officer said.

He added that tanks are also evolving to meet

emerging threats through lighter designs, anti-drone protection, electronic warfare jammers and hard-kill active protection systems capable of intercepting missiles and drones.

The Russia-Ukraine war
The Russia-Ukraine war has highlighted both the strengths and vulnerabilities of tanks. Numerous battlefield videos have shown tanks being destroyed by inexpensive drones, exposing their susceptibility in an increasingly transparent battlefield dominated by aerial surveillance and precision strikes. Although tanks can no longer spearhead assaults without support, they remain highly effective when employed as part of an integrated combined-arms operation involving infantry, artillery, air defence, electronic warfare and drones.

Recent conflicts have demonstrated that low-cost drones can impose disproportionate costs on technologically superior militaries.

During the U.S.-Iran conflict, reports indicated that U.S. Terminal High Altitude Area Defense radar sites in West Asia were targeted by inexpensive Shahed-type drones.

India witnessed a similar trend during Operation Sindoor, when Pakistan reportedly employed swarms of low-radar cross-section drones to saturate Indian airspace, overwhelm air defence systems and exhaust expensive interceptor missiles. Cheap

Why tanks still matter in modern warfare

Drones, missiles and precision-guided weapons have transformed the modern battlefield, but military experts argue they cannot achieve the ultimate objective of war: controlling territory. They say tanks and ground forces remain indispensable despite the rapid evolution of warfare



In June this year, Prime Minister Narendra Modi visited Larsen & Toubro's Hazira facility in Gujarat, where he reviewed the Zorawar light tank and other indigenous defence systems. SPECIAL ARRANGEMENT

drones costing only a few lakh rupees can force defenders to launch interceptors worth several crores. Warfare is increasingly becoming a battle of mathematics, where defenders can lose economically despite winning tactically.

Zorawar light tank

Recognising the changing character of warfare, India is developing the indigenous Zorawar light tank specifically for operations along the Line of Actual Control (LAC). In June this year, Prime Minister Na-

rendra Modi visited Larsen & Toubro's Hazira manufacturing facility in Gujarat, where he reviewed the Zorawar light tank and other indigenous defence systems.

Jointly developed by the Defence Research and De-

velopment Organisation and L&T Defence, the 25-tonne, air-transportable combat vehicle is designed for rapid deployment in high-altitude regions.

The Indian Army now expects to induct the Zorawar between 2028 and 2029. Now-retired Army Chief General Upendra Dwivedi recently stated that issues identified during development and testing are being addressed through routine design refinements. The Army plans to induct 354 Zorawar tanks under a programme estimated to cost around ₹17,500 crore. The Zorawar has been specifically designed for operations along the LAC, where China's People's Liberation Army has already deployed Type-15 light tanks. Indian military planners view the platform as critical to narrowing the operational gap and enabling rapid reinforcement of forward positions in mountainous terrain.

The T-72 Ajeya remains the Army's largest tank fleet, with approximately 2,400-2,500 tanks, many of which are over four decades old. The more capable T-90S Bhishma, numbering more than 1,200, forms the backbone of India's armoured corps.

Unique challenge

India faces a unique strategic challenge: preparing simultaneously for conventional warfare against Pakistan in the plains and armoured operations against China in the high Himalayas. The western theatre is ideal for traditional ar-

moured manoeuvre warfare. The northern theatre presents an entirely different environment, characterised by extreme altitude, narrow valleys, weak bridges and steep gradients.

At high altitudes, tank engines lose significant power because of thin air, fuel consumption increases, movement becomes difficult and recovering disabled vehicles is extremely challenging. Without a credible offensive armoured capability in the mountains, Indian planners feared continued vulnerability to incremental Chinese military pressure.

'Not disappearing'

Lieutenant General (Retd.) Dushyant Singh, director general of the Centre for Land Warfare Studies (CLAWS), said no single template applies to every conflict. "In the India-Pakistan context, territory remains central. Likewise, the Iran-U.S. conflict demonstrates that unless territory is physically occupied, victory remains incomplete. Air strikes and missiles are effective for punitive action and destroying military assets, but controlling land requires boots on the ground. Legacy platforms such as tanks remain relevant despite advances in drones and missiles," he said.

According to him, tanks now require dedicated anti-drone protection and integrated air defence support. Singh emphasised that future warfare will increasingly rely on integrat-

ed multi-domain operations involving the Army, Air Force and Navy, supported by cyber warfare, economic measures and maritime power.

'Role not ended'

Globally, tanks no longer dominate battlefields in the way they did between the Second World War and the end of the Cold War.

India is introducing an entirely new class of tanks because its principal armoured challenge lies not on drone-saturated plains but in the high Himalayas, where altitude, mobility and transportability determine operational success.

The Zorawar is designed to address a uniquely Indian strategic requirement that many Western militaries do not face. The tank is no longer the undisputed king of the battlefield. But it remains something even more enduring: an indispensable component of combined-arms warfare.

In a battlefield increasingly dominated by drones, guided missiles and precision weapons capable of destroying multi-million-dollar armoured vehicles within seconds, tanks continue to provide one capability that no drone can replicate: the ability to physically seize, occupy and hold ground. Ultimately, wars are won not only by destroying the enemy but by controlling territory. And for that, boots on the ground, supported by tanks, remain indispensable.

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India's Two-Front Armoured Challenge

Western Theatre

Plains : Pakistan

Conventional, armoured manoeuvre warfare; tanks operate at scale across open terrain

Northern Theatre

High Himalayas : China

Extreme altitude, narrow valleys, weak bridges, steep gradients; engines lose power, movement is difficult, recovery of disabled vehicles is hard

Why Tanks Still Matter in Modern Warfare

- **History**

- The need for a light tank (weighing maximum 25 tonnes) **emerged during the 2020 standoff with China in Eastern Ladakh, highlighting the requirement for lightweight, easily deployable tanks.**
- It was named after a military general, **Zorawar Singh Kahluria**, who had served under Jammu's Raja Gulab Singh in the 19th Century.

- **Features**

- It is weighing a **maximum 25 tonnes**, is a key requirement of the light tank which ensures it can be **transported by air.**
- The **tank will be able to fire at high angles of elevation** and perform the role of limited artillery and is an agile weapon platform.
- It is designed to operate in **varying terrain, from high-altitude areas and marginal terrains to island territories.**
- It will be equipped with all the modern technologies, such as **Artificial Intelligence, Drone Integration, a high degree of situational awareness, and amphibious operation capabilities**

Why Tanks Still Matter in Modern Warfare

- **Other Tanks**

- **T-90 Bhishma**

- Russian-origin third-generation Main Battle Tank (MBT).
- Equipped with a 125 mm smoothbore gun, thermal sights, and reactive armour.
- Highly advanced fire control system with day-night capabilities.
- Known for superior mobility and combat efficiency.
- Over 1,200+ units in service; produced under license in India.

- **Arjun MBT Mk-1**

- India's first indigenous third-generation MBT, developed by DRDO.
- Armed with a 120 mm rifled gun, computerised fire control, and advanced composite armour.
- Designed for high protection, firepower, and crew comfort.
- Deployed in western desert regions, around 124 units were inducted.
- Arjun MBT Mk-1A is Upgraded version of Arjun Mk-1 with 72 improvements.

- **T-72 Ajeya**

- A Soviet-origin tank in Indian service since the 1980s.
- The backbone of the Indian tank fleet is with over 2,400 units.
- Armed with a 125 mm smoothbore gun and upgraded with thermal sights and ERA.
- Currently undergoing engine and systems upgrade to extend service life.

Why Tanks Still Matter in Modern Warfare

- **The Debate**

- Drones, missiles and precision-guided weapons have transformed the battlefield, but military experts argue tanks remain indispensable for the ultimate objective of war: controlling territory

- **Lessons from Recent Conflicts**

- The Russia-Ukraine war exposed tank vulnerability to inexpensive drones; India saw a similar trend during Operation Sindoor, where low-cost drones saturated airspace

- **Induction Timeline**

- The Army expects to induct 354 Zorawar tanks between 2028 and 2029 under a programme estimated to cost around Rs 17,500 crore

- **The 'Role Not Ended' Argument**

- Tanks no longer dominate battlefields as in the WWII/Cold War era, but remain the only capability to physically seize, occupy and hold ground; future warfare requires integrated multi-domain operations

Why Tanks Still Matter in Modern Warfare

Practice Question

With reference to India's indigenous defence platforms, consider the following statements:

1. The Zorawar light tank is being jointly developed by the DRDO and L&T Defence.
2. The Zorawar tank is primarily designed for high-altitude operations along the Line of Actual Control.
3. The T-90S Bhishma is currently the largest tank fleet in the Indian Army's armoured corps.

Which of the statements given above is/are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

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UPSC Prelims 2026 Question:

Which of the following items of defence hardware is/are manufactured in India?

1. Su-30 MKI Fighter Jets
2. T-90 MK-III Tanks
3. Akula Class Submarine

Select the answer using the code given below:

1. 1 and 2

2. 1 and 3

3. 1 only

4. 2 only

India's Semiconductor Push Will Create Countless Jobs: PM

India's semiconductor push will create countless jobs: PM

Modi inaugurates semiconductor facility in Gujarat's Sanand; he says such clusters will bring in economic transformation; he urges youth to tap new opportunities in era of Artificial Intelligence

Abhinav Deshpande
AHMEDABAD

Prime Minister Narendra Modi on Saturday said India is giving birth to semiconductor clusters across the country, and described the current phase as the beginning of large-scale employment and economic transformation.

He said the country is building the entire electronics value chain, from products and components to semiconductors, which will serve as the road map for Viksit Bharat.

He said semiconductor clusters were emerging not only in Sanand (Gujarat) but at multiple locations across India.

Mr. Modi, who inaugurated the CG Semi **Out-sourced Semiconductor Assembly and Test (OSAT)** facility in Sanand, also said India's youth would power the artificial intelligence, robotics and next-generation technology revolution with made-in-India chips.

He said the plant reflected the government's vision of 'Design in India' and 'Make in India' and noted that he had laid the foundation stone for the project in 2024, chip testing began



Future ready: Prime Minister Narendra Modi at the semiconductor facility in Sanand, Gujarat, on Saturday. VIJAY SONEJI

in August 2025, and commercial production had now commenced.

Appeal to youth

Mr. Modi said the coming era of artificial intelligence would open up new avenues for skills and expertise. "The youth of India should not lose this opportunity. I want to tell the youngsters of the country: 'Your ideas, my support,'" Mr. Modi said.

"It (semiconductor industry) is the next step in the electronics revolution that has taken place in India over the past decade," he added as he was handed the first semiconductor chips made at the company, to be exported to Ja-

pan. First, it was the IT revolution that created opportunities for lakhs of Indians to showcase their talent, he said. Then the smartphone and electronics manufacturing boom opened new avenues for employment and innovation. "Now, the world is entering the era of semiconductors and artificial intelligence revolution that will create opportunities for countless people."

Government's goal

The CG facility is part of the India Semiconductor Mission and will undertake the assembly, packaging and testing of semiconductor chips. Mr. Modi said the project was built in part-

nership with companies from Thailand and Japan, and described it as a model of technology, trust and collaboration that would strengthen India's semiconductor sector.

He also said that the government's goal is to build a complete semiconductor ecosystem in India, from chip design to fabrication and packaging. "I'm informed that this unit has the capacity to manufacture 20 crore chips annually and that the company aims to scale up production to more than 500 crore chips a year, which means 1.5 crore chips per day," he said.

He called the project a "success story" and a "revolution".

He said many women employees from the hinterlands of the country who completed their ITI training were now working at the plant. He said many of them had never visited Delhi or Mumbai and had never held a passport before travelling to Malaysia for training.

"They learnt advanced semiconductor manufacturing techniques there and are now part of India's chip manufacturing sector."

India's Semiconductor Push Will Create Countless Jobs: PM

- Semiconductors are materials which have conductivity between conductors (full conductors of heat and electricity) and insulators (bad conductors of heat and electricity).
- They form the intricate circuits that power our digital world and drive technological innovations.
- Semiconductors can be pure elements, such as Silicon (second most common element in Earth's crust after Oxygen) or Germanium, or compounds like Gallium arsenide or Cadmium selenide.
- East Asian countries like Taiwan (TSMC), South Korea (Samsung), and China (NAURA Technology), which account for over 70% of the global market share.
- The United States is home to chip design and R&D tech giants like Intel and Qualcomm.
- Top Producer : Taiwan > South Korea
- Top Consumer : China > USA

India's Semiconductor Push Will Create Countless Jobs: PM

Production Linked Incentive (PLI) Scheme

MeitY notified the Production Linked Incentive (PLI) Scheme as part of the vision of NPE 2019.

Includes 4-6% domestic production-linked incentives to electronics companies that manufacture mobile phones and specified components, including Assembly, Testing, Marking, and packaging units.

It offers incentives based on incremental sales designed to boost domestic manufacturing & attract large investments in the sector.

MeitY has set up the India Semiconductor Mission (ISM) as a dedicated body to drive the program and facilitate investments.

India-US Initiative on Critical and Emerging Technologies (iCET)

In 2022, India signed the India-US iCET deal to elevate and expand Indo-U.S. strategic technology partnership and defence industrial cooperation between the countries' governments, businesses, and academic institutions.

Design-Linked Incentive (DLI) Scheme

Financial and infrastructure support will be provided to companies across various stages of development and deployment of semiconductor design for Integrated Circuits (ICs), Chips, Systems on Chips (SoCs), Systems and IP Cores, and semiconductor-linked design for over five years.

India's Semiconductor Push Will Create Countless Jobs: PM

- **Proposed Plants on India**

- **Micron Technology : Sanand, Gujarat**
- **CG Power & Industrial Pvt Ltd in partnership with Renesas and Stars : Sanand, Gujarat**
- **Tata Electronics (TEPL) in partnership with Powerchip Semiconductor Manufacturing Corp (PSMC) of Taiwan : Dholera, Gujarat**
- **Kaynes Semicon Pvt Ltd : Sanand, Gujarat**
- **Tata Semiconductor Assembly and Test Pvt Ltd (TSAT) : Morigaon, Assam**
- **HCL-Foxconn JV : Jewar, Uttar Pradesh**
- **3D Glass Solutions Inc. : Bhubaneswar, Odisha**
- **CDIL (Continental Device) : Mohali, Punjab**
- **ASIP (Advanced System in Package Technologies) : Andhra Pradesh**

India's Semiconductor Push Will Create Countless Jobs: PM

- **The Inauguration**

- PM Modi inaugurated the CG Semi Outsourced Semiconductor Assembly and Test (OSAT) facility in Sanand, Gujarat, under the India Semiconductor Mission

- **Capacity and Scale**

- The Sanand unit can manufacture 20 crore chips annually, with plans to scale up to more than 500 crore chips a year, about 1.5 crore chips per day

- **Complete Value Chain Vision**

- Government aims to build a full semiconductor ecosystem in India, from chip design and fabrication to packaging, aligned with 'Design in India' and 'Make in India'

- **Global Partnerships and Youth Impact**

- Facility built in partnership with companies from Thailand and Japan; many women employees from rural India, trained in Malaysia, now work in advanced chip manufacturing

- **Strategic Framing**

- PM described the semiconductor and AI revolution as the next phase after the IT and smartphone/electronics manufacturing booms, urging youth to seize new opportunities

India's Semiconductor Push Will Create Countless Jobs: PM

QUESTION

GS

Hard

Science & Technology

Prelims 2026

Which one of the following pairs of semiconductor plants in India and their locations is **not** correctly matched?

Semiconductor Plant	Location
(a) CG Power and Industrial Solutions Pvt. Ltd. in partnership with Renesas Electronics and STARS Microelectronics	Gujarat
(b) Tata Semiconductor Assembly and Test Pvt. Ltd.	Assam
(c) HCL-Foxconn Joint Venture India Chip Ltd.	Madhya Pradesh
(d) SicSem Pvt. Ltd.	Odisha

Thank You